

**Assignment 10: Explainable Artificial Intelligence (XAI): Concepts, Taxonomies,
Opportunities, and Challenges towards responsible AI**

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IMT 574: Data Science II : Machine Learning

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December 07, 2025

1. Why would you advocate for bringing transparency or explainability in an ML/AI system? What are the advantages to the users as well as the organization providing that system? [3 points]

- Transparency is the passive characteristic of the model that helps the human user understand the model while explainability is an active characteristic of the model that clarifies or details the internal processes of the model. One should advocate for bringing both to a ML/AI system because they help the users understand what the model is showing and interpret the results of model's output. Without these features or even low values of the features the users have a more difficult time interpreting what the model has conveyed. Both features of a model are based on the concept of understandability which considers the model users' cognitive capabilities and their needs.
- Per the article, "Explainability is one of the main barriers AI is facing now adays in regard to its practical implementation" (Barredo Arrieta et al. 2020). If users can understand not only the output but the interworking's of the ML system, they can determine which factors contribute the most to model performance to increase the accuracy of the model. This lack of understanding will limit the trust in the models and can hinder their deployment.
- For users, transparency and explainability increase trustworthiness and confidence, it can help identify potential biases and allow the user to understand how the data has influenced the outcomes. By understanding the variables and causal relationships within the system the organizations can improve the models.

2. What is a "fair" system? Take an example of an ML/AI system and tell us what achieving "fairness" would look like. [3 points]

- A fair system is one that detects potential biases in the data and accounts for the bias in the results of the data set. These biases will include religion, gender, race, sexual orientation, disability, ethnicity, or any personal condition. Fairness is not limited to removing these attributes from the dataset but rather ensuring that the system's decisions do not systematically disadvantage any specific subgroups.
- An example of a ML/AI system would be a resume filtering model that reviews all the resumes that are submitted for a specific role. To ensure that this system is 'fair' the system could consider not allowing the model to use attributes such as gender, religion, name, etc. into the training of the data to make sure that the model doesn't not have any preconceived biases to differentiate which resumes fit the role best. Rather the system should only be allowed to review the content of the resume, not even their names should be considered in the model decision making. To go one step further, the system could require continuous monitoring and implementing additional strategies such as apply bias-reduction techniques during model training. This would confirm that the system is avoiding discrimination and promoting equal treatment of all applicants.

3. How do you see concepts like bias, fairness, accountability, and transparency-related? [2 points]

- These concepts are an interconnected concept that together form the foundation for responsible AI. Bias represents the systemic error or skewed patterns in data that can lead to discriminatory outcomes. Fairness is the attribute that works to deter discriminatory effects by ensuring that no group is unjustly disadvantaged by the model's outcomes. Transparency allows the users to understand how the decisions are being made by showing the logic and processes used within the model. This understanding allows the users to review potential biases and evaluate their effect. Without transparency detecting these biases, it can be difficult to understand where the biases lie. Accountability ensure that individuals and organizations are responsible for the impacts of the AI systems and are responsible for any potential harm or discrimination that may occur.
- All together, these concepts operate in a cyclical patten where the transparency exposes the bias, accountability requires corrective action, and fairness continues to measure and manage the predetermine biases.

4. Give an example of a system that is biased but fair. [2 points]

- A system that is biased and fair would be a system designed to hire construction workers for an upcoming project that is very physically intensive. There is an inherent bias against older people, women, and applicants with less training because the role requires a more significant physical capability and technical skill. However, this bias can only be considered fair when it is directly related to legitimate job requirements necessary to ensure the safety and efficiency of the workers during the duration of the project.
- To maintain fairness, the hiring process must be consistent across every hire regardless of gender or other personal characteristics. For example, if a female applicant applies for the position they should be interviewed and evaluated at the same standards as a male applicant would be. If she demonstrates she is capable to perform every job requirement they should be given the role to ensure fair hiring practices. This ensures that while the system prioritizes job-relevant capability, it still provides equal opportunity and upholds fair hiring process in a biased field.

Resources

Barredo Arrieta, A., Díaz-Rodríguez, N., Del Ser, J., Bennetot, A., Tabik, S., Barbado, A., García, S., Gil-Lopez, S., Molina, D., Benjamins, R., Chatila, R., & Herrera, F. (2020). Explainable Artificial Intelligence (XAI): Concepts, taxonomies, opportunities and challenges toward responsible AI. *Information Fusion*, 58, 82-115.